

LINUX PROGRAMMING

III B. TECH- II SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS16	PCC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

The course should enable the students to:

1. Understand basic principles of Linux Internals.
2. To learn Linux process control and shell programming.
3. Explain the basic Methods on which the Linux kernel is built upon.
4. To facilitate students in understanding process.
5. To facilitate students in understanding Inter process communication.

COURSE OUTCOMES:

After completion of the course the student is able to

1. Understand how to work with Linux commands and how to write Shell Scripts.
2. Apply fundamental knowledge of Linux Internals in Real time scenarios.
3. Demonstrate tools and interfaces to successfully develop new features of the kernel.
4. Design client server application to support communication interfaces.

UNIT-I	INTRODUCTION & LINUX UTILITIES	
Linux Utilities: File handling utilities, Security by file permissions, process utilities, Disk utilities, networking commands, filters, text processing utilities, back up utilities. Sed- scripts, operation, addresses, commands, applications, awk- execution, fields and records, scripts, operations, patterns, actions, functions, using system commands in awk. Working with bourne again shell(bash): Introduction, shell responsibilities, pipes and redirection, output redirection, here documents, running a shell script, the shell as programming language, shell meta characters, file name substitution, shell variables, command substitution, shell commands, the environment, quoting, test command, control structures, arithmetic in shell, shell script examples, interrupt processing, functions, debugging shell scripts		
UNIT-II	FILES	
Files: File Concept, File System Structure, Inodes, File Attributes, File types, Library functions, the standard I/O and formatted I/O in C, stream errors, kernel support for files, System calls, file descriptors, low level file access – File structure related system calls(File APIs), file and record locking, file and directory management – Directory file APIs, Symbolic links & hard links		
UNIT-III	PROCESS & SIGNALS	
Process: Process concept, process identifiers, process structure, process table, viewing processes, system processes, process scheduling, Process creation: waiting for a process, process termination, zombie processes, orphan process, process APIs. Signals: Introduction to signals, signal generation and handling, kernel support for signals, signal function, unreliable signals, reliable signals, kill, raise, alarm, pause, abort, sleep functions.		
UNIT-IV	IPC	
Inter Process Communication: Introduction to IPC, IPC between processes on a single computer system, IPC between processes on different systems, pipes-creation, IPC between related processes using unnamed pipes, FIFOs-creation, IPC between unrelated processing using FIFOs(named pipes), differences between unnamed and named pipes, popen and pclose library functions. Message queues: kernel; support for message queues, unix system V APIs for messages, client/server example.		

Semaphores- kernel support for semaphores, unix system V APIs for semaphores, file locking with semaphores.		
UNIT-V	SOCKETS	
Sockets: Introduction to Berkeley Sockets, IPC over a network, Client-Server model, Socket address structures (Unix domain and Internet domain), socket system calls for connection oriented protocols and connectionless protocol, Example-client/server programs-Single server-client connection, Multiple simultaneous clients, Socket options-setsocket and fcntl system calls, Comparison of IPC mechanisms		
TEXT BOOKS:		
1. W. Richard. Stevens (2005), Advanced Programming in the UNIX Environment, 3rd edition, Pearson Education, New Delhi, India. 2. Unix and shell Programming Behrouz A. Forouzan, Richard F. Gilberg.Thomson		
REFERENCE BOOKS:		
1. Linux System Programming, <i>Robert Love, O'Reilly, SPD.</i> 2. Advanced Programming in the UNIX environment, 2nd Edition, <i>W.R.Stevens</i>, Pearson Education. 3. UNIX Network Programming, <i>W.R. Stevens</i>, PHI. 4. UNIX for Programmers and Users, 3rd Edition, <i>Graham Glass, King Ables</i>, Pearson Education 5. B. A. Forouzan, R. F. Gillberg, "C Programming and Data Structures", Cengage Learning, India,		
WEB REFERENCES:		
1. https://www.bfoit.org/itp/Programming.html 2. https://www.khanacademy.org/computing/computer-programming 3. https://www.edx.org/course/programming-basics-iitbombayx-cs101-1x-0 4. https://www.edx.org/course/introduction-computer-science-harvardx-cs50x		
E-TEXT BOOKS:		
1. http://www.freebookcentre.net/Language/Free-C-Programming-Books-Download.htm 2. http://www.imada.sdu.dk/~svalle/courses/dm14-2005/mirror/c/ 3. http://www.enggnotebook.weebly.com/uploads/2/2/7/1/22718186/ge6151-notes.pdf		
MOOC COURSE		
1. https://www.alison.com/courses/Introduction-to-Programming-in-c 2. http://www.ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-s096-effective-programming-in-c-and-c-january-iap-2014/index.htm		